a) $T\hat{j} - mg\hat{j} = ma\hat{j}$ b) $\vec{T} + \vec{F} = \vec{A} + \vec{B}$

c) $\vec{T} - \vec{F} = \vec{A} - \vec{B}$ d) $T\hat{j} + mg\hat{j} = ma\hat{j}$

[Ans: (a) $T - mg = ma$

(b) $T_x + F_x = A_x + B_x$ etc.]

14. Calculate the area of the triangle for which two of its sides are given by the vectors $\vec{A} = 5\hat{i} - 3\hat{j}$, $\vec{B} = 4\hat{i} + 6\hat{j}$

[Ans: Area = 21 squared units]

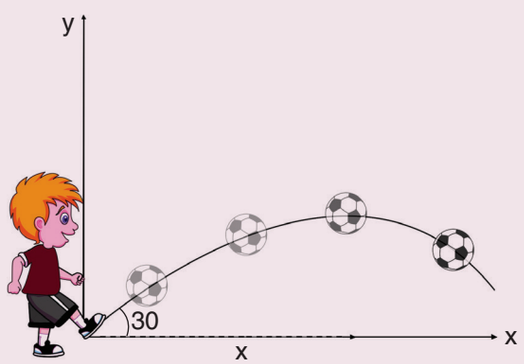
15. If Earth completes one revolution in 24 hours, what is the angular displacement made by Earth in one hour. Express your answer in both radian and degree.

[Ans: $\theta = 15° \; or \; \frac{\pi}{12}$]

16. A object is thrown with initial speed 5 m s^{-1} with an angle of projection 30°. What is the height and range reached by the particle?

[Ans: height = 0.318 m Range =2.21 m]

17. A foot-ball player hits the ball with speed 20 m s^{-1} with angle 30° with respect to horizontal direction as shown in the figure. The goal post is at distance of 40 m from him. Find out whether ball reaches the goal post?



[Ans: Ball will not reach the goal post. The range =35.3 m]

18. If an object is thrown horizontally with an initial speed 10 m s^{-1} from the top of a building of height 100 m. what is the horizontal distance covered by the particle?

[Ans: R= 45 m]

19. An object is executing uniform circular motion with an angular speed of $\frac{\pi}{12}$ radian per second. At t = 0 the object starts at an angle θ = 0 What is the angular displacement of the particle after 4 s?

[Ans: 60°]

20. Consider the x–axis as representing east, the y–axis as north and z–axis as vertically upwards. Give the vector representing each of the following points.

a) 5 m north east and 2 m up

b) 4 m south east and 3 m up

c) 2 m north west and 4 m up

[Ans: (a) $\frac{5(\hat{i} + \hat{j})}{\sqrt{2}} + 2\hat{k}$ (b) $\frac{4(\hat{i} - \hat{j})}{\sqrt{2}} + 3\hat{k}$

(c) $(-\hat{i} + \hat{j})\sqrt{2} + 4\hat{k}$]

21. The Moon is orbiting the Earth approximately once in 27 days, what is the angle transversed by the Moon per day?

[Ans: 13°3']

22. An object of mass m has angular acceleration $\alpha = 0.2 \, rad \, s^{-2}$. What is the angular displacement covered by the object after 3 second? (Assume that the object started with angle zero with zero angular velocity).

[Ans: 0.9 rad or 51°]

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